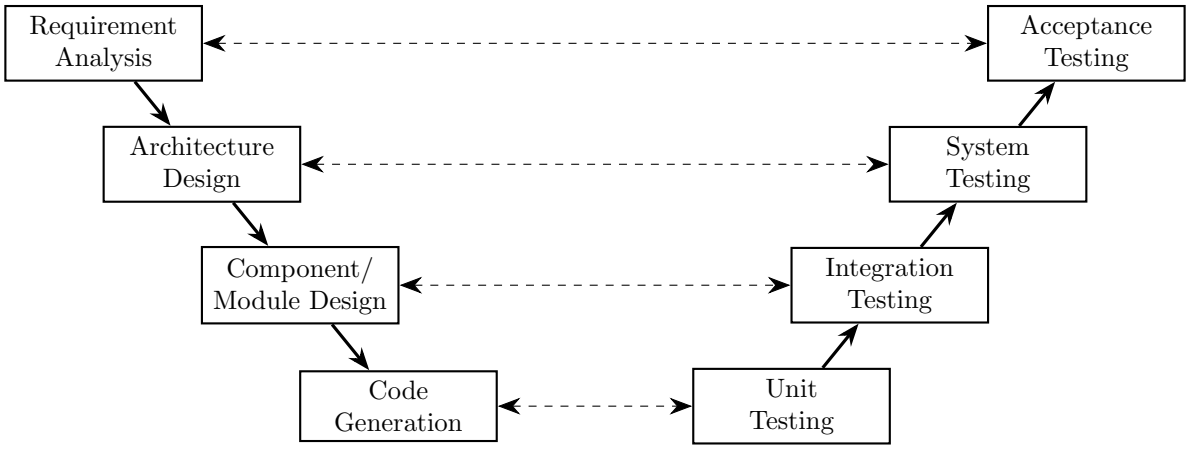
Quiz 1

CSE 470: Software Engineering

1.

Draw and label the V Model. (5 marks)



2.

A banking transaction management system is being developed for a national bank. The system will handle customer accounts, deposits, withdrawals, loan processing, and fund transfers. Due to financial regulations and security requirements, all system requirements are fully defined and documented before development begins. The bank requires thorough verification and validation at every stage to ensure accuracy, reliability, and compliance. Any defects must be identified as early as possible because errors could lead to financial losses and legal issues. (5 marks)

Suitable model: V-Model.

- Requirements are **fully defined upfront**: the V-Model, like Waterfall, assumes a fixed, well-documented requirement set before design starts, which fits a regulated banking domain.
- The V-Model pairs **every development phase with a corresponding testing phase** (requirements ↔ acceptance testing, design ↔ system/integration testing), giving the thorough verification and validation at every stage that the bank needs.
- Test cases are planned **in parallel with each development stage**, so defects tied to requirements or design are caught **early**, before they propagate. This is critical when errors mean financial loss or legal liability.

3.

A Virtual Reality (VR) Fitness Training Platform is being developed. The application allows users to participate in immersive workout sessions using VR headsets. The first version will include basic exercise routines and performance tracking. After each release, users will give review on workout quality, motion tracking accuracy, user interface design, and overall experience. Based on this feedback, the development team plans to refine existing features, improve user engagement, enhance AI-based coaching, add multiplayer workout sessions, and optimize the VR environment over several development cycles. Which software development model would be most suitable for this project? Support your answer with proper logic. (5 marks)

Suitable model: Iterative Model.

- The project starts with a **basic working version** (core exercise routines and tracking) and features are refined over time. This is the defining trait of the Iterative Model, unlike Waterfall/V-Model which need complete upfront requirements.
- **User feedback after each release** (motion tracking, UI, engagement) directly drives what is refined in the next cycle, matching the Iterative Model's build, evaluate, refine loop.
- New features (AI coaching, multiplayer) are **added step-by-step over several cycles** rather than specified all at once, which the Iterative Model accommodates but a sequential model does not.